

Topic Module Computer Laboratory	Hydraulic Modelling	
	1.	Surface Water Modelling
	3	Flood assessment for different AEPs pre- and post-implementation of a bridge with culverts

You are tasked to support the assessment of the potential impact of flood events of varying annual exceedance probability both pre- and post-implementation of a bridge with culverts. You have been provided with the river geometry data and information regarding the planned culvert.

Design brief: Provide an assessment of the effect of flood events of varying magnitude and the construction of a bridge with culverts on the water surface profile.

Tasks

(i) Determine the water surface profile for the flood events with an annual exceedance probability of 20% (5 year return period) ($8 \text{ m}^3/\text{s}$), 10% (10 year return period) ($12 \text{ m}^3/\text{s}$) and 4% (25 year return period) ($17 \text{ m}^3/\text{s}$). (ii) Determine the water surface profile for the three flood events, but after a bridge with culverts has been installed in the river. Employ steady flow analysis throughout for this initial assessment. The river geometry data pre-installation of the bridge with culverts are given (file river.g01).

Part (i)

Set up the project:

File – New Project ...

Title: Water surface profile

File Name: Watersurfaceprofile.prj

Elements of a project:

- Geometric Data
- Flow Data Steady/Unsteady (Flow Data and Boundary Conditions)
- Plan (Flow Analysis)

Unit System

Change to Metric Unit System:

Options – Unit system

and activate **Set as default for new projects**

Geometric Data

Edit – Geometric Data

File –New Geometric Data

Title: River System

Import River System:

File – Import Geometric Data - HEC-RAS Format

Title: River

Filename: river.g01

Import data as: SI metric units

Button **Finished – Imported Data**

Adjust view:

View – Zoom in

Enlarge to Window Extent

View – Set Schematic Plot Extents

Button **Set to Current View**

File – Save Geometric Data

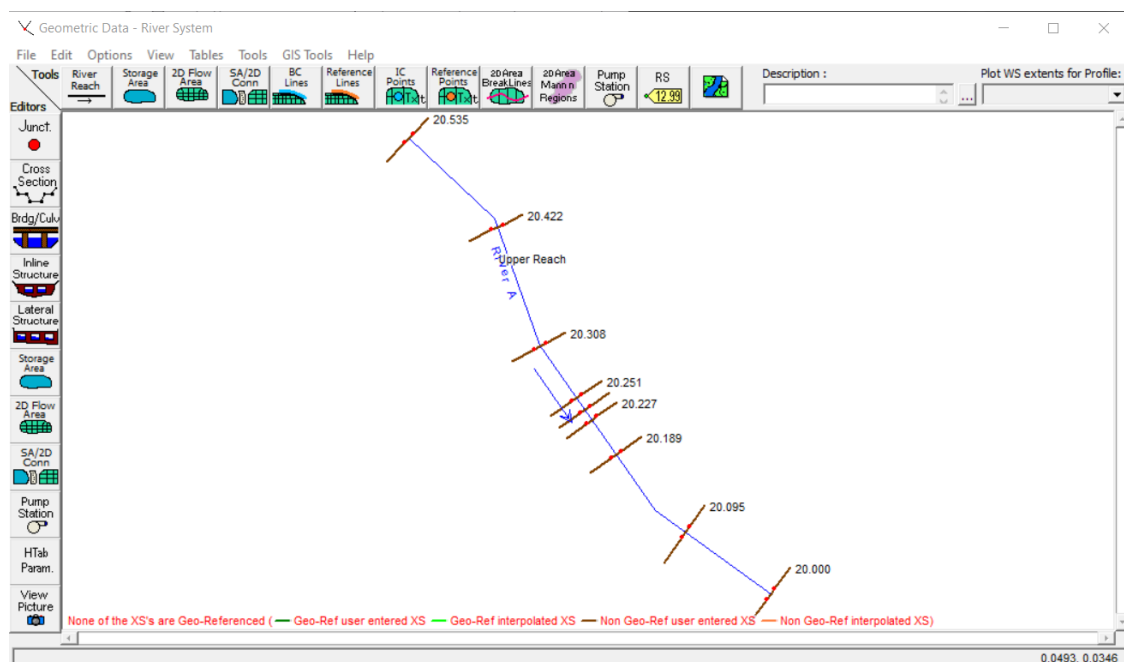


Figure 1 Geometric Data

Steady Flow Data

To perform a steady flow analysis, flow data and boundary conditions have to be specified.

Edit – Steady Flow Data

Flood Data

Number of Profiles:

Enter / Edit Number of Profiles (2000 max): 3

Rename Profiles:

Options – Edit Profile Names

Profile 1: 5 yr Flood

Profile 2: 10 yr Flood

Profile 3: 25 yr Flood

Steady Flow Data

File Options Help

Description : Flow data

Enter/Edit Number of Profiles (32000 max): 3 Reach Boundary Conditions ...

Locations of Flow Data Changes

River: River A Add Multiple...

Reach: Upper Reach River Sta.: 20.535 Add A Flow Change Location

Flow Change Location				Profile Names and Flow Rates		
	River	Reach	RS	5 yr Flood	10 yr Flood	25 yr Flood
1	River A	Upper Reach	20.535	8	12	17

Edit Steady flow data for the profiles (m3/s)

Figure 2 Flow Data

Boundary Conditions:

Subcritical flow regime → downstream boundary condition required

Boundary Condition: Known Water Surface (W.S.) can for example be extracted from the rating curve or based on historical observations.

HEC-RAS

Set known water surfaces for flows.

	Flow (m ³ /s)	Known WS El (m)
1	8	9.1
2	12	9.5
3	17	9.7

OK Cancel

Figure 3 Entering the Boundary Conditions

File – Save Flow Data As ...

Title: WS Flow Data

Calculation

Run – Steady Flow Analysis

File New Plan

(**Title:** WS Calculation 1, **Short Plan Identifier:** Calc 1)

Flow Regime: subcritical

Calculate the Critical Depth:

activate **Options – Critical Depth Output Option**

Button **Compute**

After calculation check Error Warnings

View – Summary, Err, Warn, Notes...

Viewing results

Plot (for the different floods):

Water surface: **View – Water Surface Profiles**

Cross Sections: **View – Cross Sections**

Tabular output:

View – Detailed Output Tables (Type – activate Cross Section)

View – Profile Summary Table

Variation of the Input Data – sensitivity of the results on Manning's n

Vary Manning values: **Tables – Manning's n or k Values** (in the geometric data editor)

Save the Geometric Data: **File - Save Geometric Data As..**

and compare the results.

Part (ii)

Add the bridge and culvert from Figure 4 to the river from part (i) and analyse the water levels for the three floods.

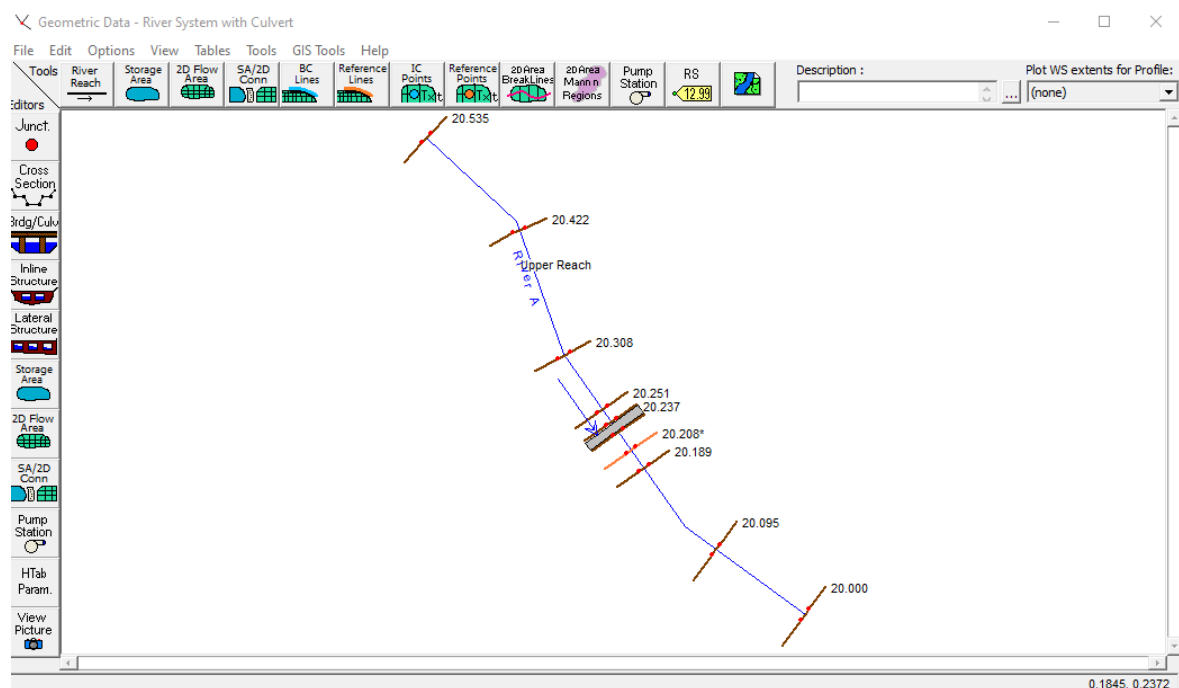


Figure 4a Bridge and Culvert Location

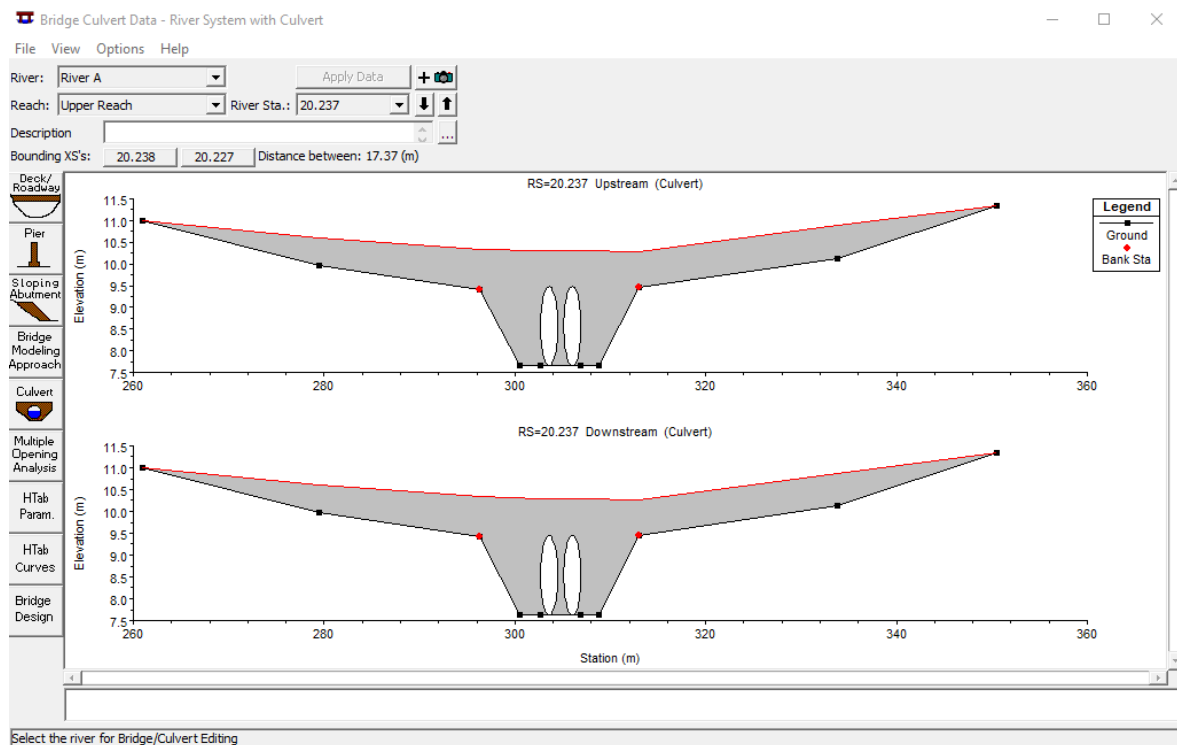


Figure 4b Bridge and Culvert Cross Section

Culvert

Add a culvert:

Icon **Bridge/Culvert**

Options – Add a bridge and/or Culvert

River station: 20.237 (Bounding stations 20.227 and 20.238)

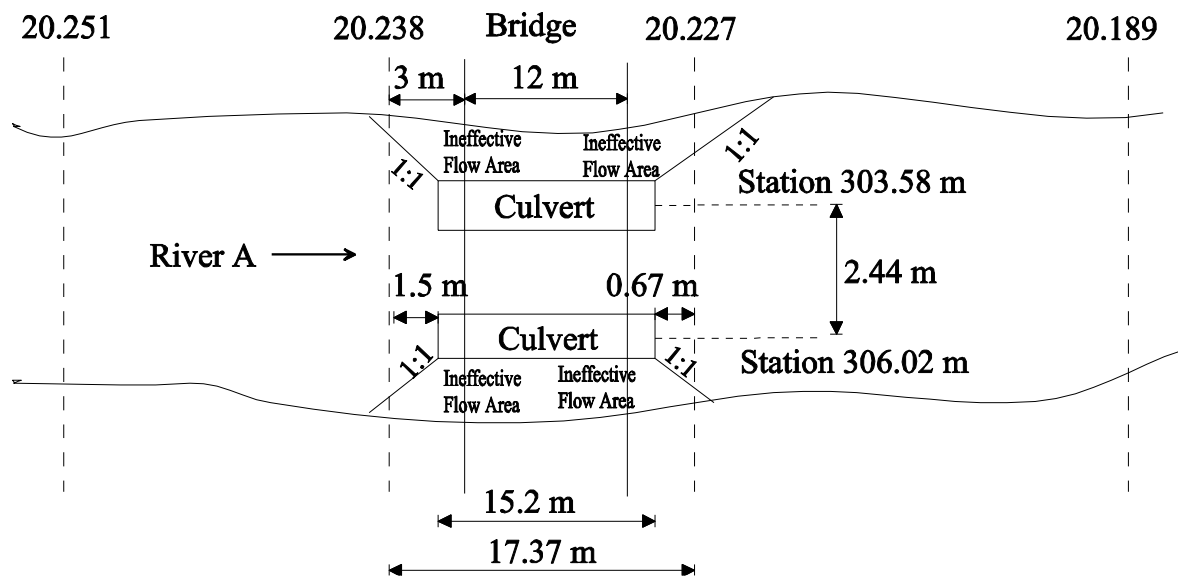


Figure 5 Culvert Layout

Deck/Roadway Data

Icon **Deck/Roadway Data**

Enter the data of the deck/roadway for Upstream

Button: **Copy US to DS** (Upstream to Downstream)

Deck/Roadway Data Editor

Distance	Width	Weir Coef
3.	12.	2.6

Clear Del Row Ins Row Copy US to DS

Upstream				Downstream			
	Station	high chord	low chord		Station	high chord	low chord
1	260.9	11		260.9	11		
2	279.5	10.61		279.5	10.61		
3	296.3	10.33		296.3	10.33		
4	302.7	10.3		302.7	10.3		
5	306.9	10.3		306.9	10.3		
6	313	10.27		313	10.27		
7	333.8	10.88		333.8	10.88		
8	350.6	11.34		350.6	11.34		

U.S Embankment SS 2. D.S Embankment SS 2.

Weir Data
 Max Submergence: 0.95 Min Weir Flow El: 10.27
 Weir Crest Shape
☒ Broad Crested
☐ Ogee

OK Cancel

Enter distance between upstream cross section and deck/roadway. (m)

Distance to cross section upstream

Crest width

Weir coefficient (Shape of the roadway, degree of the obstructions)

Upstream and Downstream Embankment Side Slopes 2 (2 horizontal to 1 vertical)

Maximum allowable submergence ratio. Ratio of downstream flow to upstream energy. When ratio exceeded, calculation will switch from weir flow equation to energy method

Minimum Weir Flow Elevation, from this elevation weir flow equation will be used

Figure 2 Deck/Roadway Editor

Culvert Geometric Data

Icon **Culvert**

Help: press for typical values

The screenshot shows the 'Culvert Data Editor' dialog box. It contains various input fields for culvert geometry and flow data. Annotations with arrows point to specific fields: 'Solution Criteria, inlet control or outlet control as the final answer for the U.S. energy grade value' points to 'Solution Criteria: Computed Flow Control'; 'Diameter of the culvert' points to 'Diameter: 1.83'; 'Upstream/ Downstream elevation of the Invert' points to 'Upstream Invert Elev: 7.65' and 'Downstream Invert Elev: 7.62'; 'Distance of the entrance of the culvert to the upstream cross section' points to 'Distance to Upstrm XS: 1.5'; 'Manning values of the top and sides / bottom of the culvert to determine the friction losses' points to 'Manning's n for Top: 0.013' and 'Manning's n for Bottom: 0.013'. A callout box at the bottom states: 'If the bottom and top n value are the same → set values zero'.

Culvert Data Editor

Culvert Group: Culvert #1

Solution Criteria: Computed Flow Control

Shape: Circular Span: Diameter: 1.83

Chart #: 1 - Concrete Pipe Culvert

Scale #: 1 - Square edge entrance with headwall

Distance to Upstrm XS: 1.5

Culvert Length: 15.2

Entrance Loss Coeff: 0.5

Exit Loss Coeff: 1

Manning's n for Top: 0.013

Manning's n for Bottom: 0.013

Depth to use Bottom n: 0

Depth Blocked: 0

Upstream Invert Elev: 7.65

Downstream Invert Elev: 7.62

Culvert Barrel Data

Barrel Centerline Stations # Barrels: 2

Barrel Name	US Sta	DS Sta
1 Barrel #1	303.58	303.58
2 Barrel #2	306.02	306.02
3		
4		
5		

Barrel GIS Data: Barrel #1 Length: 0

X	Y
1	
2	
3	
4	
5	

Individual Barrel Centerlines ... Show on Map OK Cancel Help

Select solution criteria

If the bottom and top n value are the same → set values zero

Figure 3 Culvert Data

Ineffective Flow Editor

Ineffective flow areas define areas of the cross section that will contain water, but the water is not flowing in the downstream direction. This option is typically used at cross sections in the vicinity of a culvert or bridge.

➔ Ineffective flow areas have to be defined for the cross sections immediately upstream (20.238) and immediately downstream of the culvert (20.227)

Typically, the stationing of the flow areas are set to a 1:1 ratio to the distance from the opening (see Figure 5).

Upstream station (20.238):

Distance to the opening: 1.5 m

left edge of the culverts: station: $303.58 \text{ m} - 1.83/2 \text{ m} = 302.67 \text{ m}$

right edge of the culverts: station: $306.02 \text{ m} + 1.83/2 \text{ m} = 306.94 \text{ m}$

➔ Ineffective flow areas are set at cross section 20.238 at stations $302.67 - 1.5 = 301.17 \text{ m}$ and $306.94 + 1.5 = 308.44 \text{ m}$

Elevation = 10.27 m (a value equal to the high cord on the upstream side of the crest)

Downstream station (20.227)

Distance to the opening: 17.37 m (Distance between 20.238 and 20.227) - 1.5 m (distance of the opening from 20.238) – 15.2 m (culvert length) = 0.67 m

➔ Ineffective flow areas are set at cross section 20.227 at stations $302.67 - 0.67 = 302 \text{ m}$ and $306.94 + 0.67 = 307.61 \text{ m}$

Elevation = 9.75 m (a value smaller to the high cord on the downstream side of the crest)

To add ineffective flow areas:

Options – Ineffective Flow Areas (in the Cross Section Editor, access via Cross Section in the Geometric Data Editor)

Required Cross Sections for Calculation

The culvert routine utilises four cross sections specifically located on both sides of the structure to determine the energy losses through the culvert.

First Cross Section

It should be located sufficiently downstream from the culvert so that the flow is not affected by the structure. The distance should be determined by field studies during high floods, a procedure outlined by the USACE (see documentation) or other acceptable procedures.

The distance should be 30 m (after USACE), the distance between Cross Section 20.227 and 20.189 is 60.96 m

➔ an additional cross section is needed

Tools – XS Interpolation – Between 2 Xs's (Geometric data editor)

Scroll to Cross Section 20.227 to 20.189

Maximum distance: 30.5 m

Button **Interpolate**

Second Cross Section

This section should be close to the culvert downstream. Cross Section 20.227 is situated close to the culvert.

Third Cross Section

This section should be close to the culvert upstream. Cross Section 20.238 is situated close to the culvert.

Fourth Cross Section

This section should be located upstream of the culvert where the flow lines are parallel. After the procedure outlined by the USACE (see documentation) the distance should be 24.38 m. the distance of the Cross Section 20.251 is 22.84 m. These two values are sufficiently close.

Save new geometry:

File – Save Geometric Data As ...(in the geometric data editor)

Filename: River System and Culvert

Perform a new calculation

Calculation

Run – Steady Flow Analysis

File New Plan

(**Title:** WS Calculation 2, **Short Plan Identifier:** Calc 2)

Compute

View results for the different floods (Profile Plot; X-Y-Y Perspective plot; Cross Section plot; Summary Tables)

Variation of the Input Data

Vary the input data of the culvert (e.g. diameter, loss coefficients)

Save the Geometric Data: **File-Save Geometric Data As..**

and compare the results.

Discussion

Discussion of key results.

Materials

The HEC-RAS documentation is an excellent resource:

<https://www.hec.usace.army.mil/confluence/rasdocs/rasum/latest>

Note the specific section on Bridges and Culverts:

<https://www.hec.usace.army.mil/confluence/rasdocs/rasum/latest/entering-and-editing-geometric-data/bridges-and-culverts>